EduIllustrate: An Agentic Pipeline for Generating Diagram-Rich Explanations of K-12 STEM Problems

Anonymous ACL submission

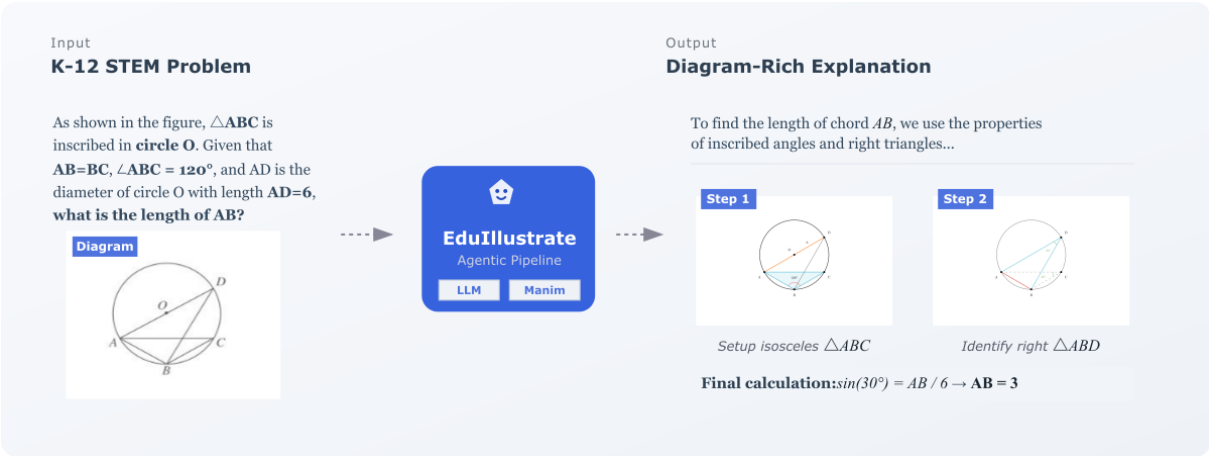

Figure 1: EduIllustrate takes a K-12 STEM problem as input and produces a structured explanation interleaved with geometrically accurate diagrams rendered via Manim. *Left*: the input problem (Grade 9 Mathematics). *Right*: two of four generated scenes illustrating the inscribed angle theorem and the resulting right triangle used to compute $AB = 3$.

Abstract

Generating high-quality illustrated explanations for K-12 STEM problems requires both geometrically accurate diagrams and coherent step-by-step reasoning—a combination that existing LLM-based educational content systems have not addressed. We present **EduIllustrate**, an automated four-stage agentic pipeline that takes a K-12 STEM problem as input and produces a structured, diagram-rich explanation with programmatically rendered static diagrams via Manim. Our pipeline introduces a sequential anchoring strategy: Scene 1 undergoes full implementation planning to establish visual conventions (color scheme, label formatting, spatial layout), while subsequent scenes generate in parallel conditioned on Scene 1's code, enforcing cross-diagram consistency without global optimization. To enable systematic evaluation, we construct a benchmark of **118 problems** spanning three grade levels (elementary, middle, high school) and five subjects (mathematics, physics, chemistry, biology, geography), and propose an **8-dimension evaluation rubric** grounded in multimedia learning theory covering both text quality and visual quality. Evaluation of five frontier LLMs shows Gemini 3.0 Pro Preview achieves the highest overall score (4.40/5.0), while Kimi-K2.5 offers the best cost-efficiency (4.07 at $0.15/problem). Workflow ablation confirms our sequential anchoring improves Visual Consistency by 12% over a fully parallel baseline at 17.5% lower cost. Human evaluation with 20 expert raters validates LLM-as-judge reliability for objective dimensions ($\rho = 0.89$ for logical coherence, $\rho = 0.83$ for diagram-problem alignment), while revealing limitations on subjective visual assessment.

1 Introduction

High-quality illustrated explanations are fundamental to K-12 STEM pedagogy. Research in educational psychology demonstrates that combining textual reasoning with visual diagrams significantly enhances conceptual understanding, particularly for abstract scientific concepts and mathe-

matical problem-solving (Mayer, 2001). However, producing such multimodal educational content at scale remains labor-intensive, requiring both deep domain expertise and technical proficiency in diagram creation tools.

Recent advances in large language models (LLMs) have enabled curriculum-aligned content generation (Liu et al., 2025), personalized learning frameworks (Wang et al., 2025), and programmatic educational video generation via Manim animations (Ku et al., 2025; Chen et al., 2025).

Yet a critical gap remains for **K-12 STEM education**: existing approaches target university-level theorems with long-form animations, while K-12 problem explanations require **static, geometrically accurate diagrams**—carrying precise semantic meaning across subjects (exact angle measurements in geometry, force vectors in physics, molecular structures in chemistry)—seamlessly integrated with step-by-step reasoning. No unified benchmark exists to jointly evaluate textual and visual quality across K-12 STEM disciplines.

We address these challenges with **EduIllustrate**, an automated four-stage pipeline generating illustrated K-12 STEM explanations with programmatically rendered diagrams via Manim. Our system decomposes generation into: (1) structured outline with interleaved text and scene specifications, (2) per-scene implementation planning with explicit spatial constraints, (3) parallel Manim code generation and rendering, and (4) final Markdown assembly. To systematically evaluate multimodal educational content, we construct a benchmark of **118 K-12 STEM problems** spanning three grade levels and five subjects, proposing an **8-dimension evaluation rubric** grounded in multimedia learning theory.

Our contributions are:

1. An automated four-stage pipeline for generating diagram-rich K-12 STEM explanations with explicit geometric accuracy and visual consistency constraints

2. A curated benchmark of 118 problems across 3 grade levels and 5 subjects for standardized multimodal evaluation

3. An 8-dimension evaluation rubric with automated and human protocols grounded in educational psychology

4. Comparative study of five frontier LLMs identifying model-specific strengths and failure modes

5. Validation of LLM-as-judge reliability for multimodal content through correlation analysis with human expert ratings

2 Related Work

2.1 LLMs in Education

LLM applications in education span intelligent tutoring, automated assessment, and adaptive content generation. Kamalov et al. (2025) reviewed agentic workflows in education, highlighting advancements in automated tutoring while noting constraints in adaptability that multi-agent frameworks address. Wang et al. (2025) introduced GenMentor, an LLM-powered multi-agent framework for goal-oriented learning, demonstrating effective skill gap identification and personalized learning path scheduling.

For K-12-specific applications, Liu et al. (2025) proposed COGENT, a curriculum-oriented framework generating grade-appropriate content aligned with curriculum standards. Despite these advances, existing work predominantly focuses on text-only explanations or assessment, leaving multimodal content generation for K-12 STEM underexplored.

2.2 Programmatic Educational Content Generation

Recent work has explored generating educational content through executable code. Ku et al. (2025) introduce an agentic approach for generating long-form theorem explanation videos using Manim animations, targeting university-level mathematics and physics. Chen et al. (2025) propose a code-centric agent framework for generating professional educational videos via executable Python code. Both systems target **video-based explanations** for advanced topics. Our work differs by focusing on **static diagram generation** for K-12 problem-solving explanations, where textbook-style illustrations enable self-paced learning across five subjects with subject-specific diagram conventions.

2.3 Programmatic Diagram Generation

Programmatic diagram generation leverages tools like TikZ, Manim, and Matplotlib. Kumar et al. (2025) introduced DiagramIR, an automatic evaluation pipeline for educational math diagrams us-

ing intermediate representations of LaTeX TikZ code, demonstrating higher agreement with human raters than LLM-as-judge baselines. Cui et al. (2025) proposed Draw with Thought, a training-free framework guiding multimodal LLMs to reconstruct scientific diagrams into editable mxGraph XML code through Chain-of-Thought reasoning.

Our work introduces **domain-specific constraints for educational diagrams**: safe margins, minimum element spacing, and discipline-specific conventions, while addressing **visual consistency across multiple diagrams** through conditional generation where subsequent scenes reference Scene 1's code as a visual anchor.

2.4 LLM-as-Judge Evaluation

LLMs as evaluators provide practical alternatives to costly human evaluation. Recent analyses show LLM-as-judge achieves 80% human agreement at 500× lower cost (EvidentlyAI Team, 2025), though systematic biases including position bias and self-preference remain. Lee et al. (2025) revealed in the legal domain that reference-free evaluation protocols correlate better with human expert judgments. Park et al. (2025) demonstrated AGACCI framework improves accuracy through distributing specialized evaluation roles across collaborative agents.

Existing LLM-as-judge frameworks focus on text quality with limited visual assessment. Our work extends this methodology to multimodal educational content, decomposing evaluation into text-only and vision-enabled dimensions and validating automated metrics against 20 human expert raters.

2.5 Multi-Modal Learning Theory

Our evaluation framework is grounded in multimedia learning theories. **Cognitive Load Theory** (Sweller, 1988) posits working memory has limited capacity—effective diagrams reduce intrinsic load by externalizing spatial relationships. **Dual Coding Theory** (Paivio, 1971) proposes information is processed through separate verbal and visual channels, with learning enhanced when both engage. **Multimedia Learning Principles** (Mayer, 2001) provide design guidelines including spatial contiguity, temporal contiguity, and coherence.

Ji et al. (2025) demonstrated that coordinating specialized agents for instructional planning, reasoning decomposition, and visualization design substantially improves educational alignment in pedagogical visualization tasks. These frameworks inform our 8-dimension rubric: Correctness & Completeness and Logical Coherence ensure cognitive accuracy; Pedagogical Effectiveness and Typographic Clarity align with coherence principles; Diagram–Problem Alignment and Text–Diagram Coordination operationalize dual coding and spatial contiguity; Element Layout Quality and Visual Consistency reduce extraneous load through predictable visual patterns.

3 Method

3.1 Task Definition

The agent receives a K-12 STEM problem statement (with an optional problem diagram image) and outputs a structured Markdown document containing step-by-step textual explanations with pedagogical scaffolding interleaved with programmatically rendered static PNG diagrams via Manim.

3.2 EduIllustrate Agent

We develop **EduIllustrate Agent**, a four-stage pipeline that decomposes illustrated explanation generation into sequential stages addressing specific multimodal content creation challenges. Figure 2 illustrates the complete pipeline.

Stage 1: Structured Outline Generation. The first stage transforms raw K-12 STEM problems into structured outlines interleaving textual explanation blocks with visual scene specifications. The model outputs an XML-based format with `<SCENE_OUTLINE>` tags containing alternating `<TEXT_k>` and `<SCENE_k>` blocks. Each `<TEXT_k>` block contains pedagogical explanation; each `<SCENE_k>` block specifies geometric/domain-specific visual elements, text annotations, color coding, and spatial layout. `<SCENE_k>` blocks are generated only when visual representations genuinely aid conceptual understanding. This structured format enables modular processing for parallel scene generation and error recovery.

Stage 2: Scene Implementation Planning. The second stage applies exclusively to Scene 1; Scenes 2–N skip this stage entirely and proceed directly to code generation. It converts high-level `<SCENE_1>` specifications into detailed implementation plans before code generation. The plan describes intended visual appearance in natural

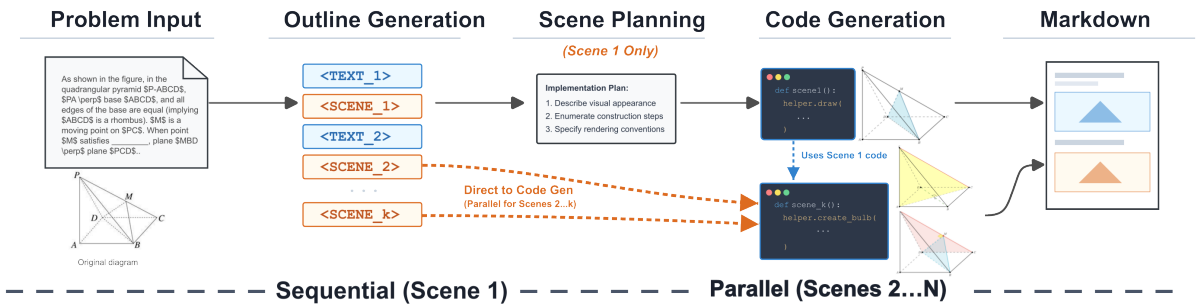

Figure 2: EduIllustrate pipeline overview. Scene 1 undergoes full sequential processing (outline → implementation planning → code generation); Scenes 2–N skip implementation planning and generate in parallel, each conditioned on Scene 1's code as a visual anchor.

language, enumerates construction steps with explicit spatial constraints (safe area bounds, minimum element spacing, relative positioning), and specifies discipline-specific rendering conventions (right-angle markers in geometry, vector arrows in physics, VSEPR molecular geometry in chemistry), establishing visual conventions that all subsequent scenes inherit.

Stage 3: Code Generation and Rendering. The third stage translates implementation plans into executable Manim code and renders static PNG images. Scene 1 undergoes fully sequential generation, establishing the visual "anchor" defining color scheme, labeling conventions, line weights, and font sizes. After Scene 1 completes, remaining scenes generate in parallel, each conditioned on Scene 1's complete Manim code to maintain visual consistency across diagrams.

Stage 4: Document Assembly. The final stage assembles <TEXT_k> blocks and rendered scene images into a structured Markdown document in alternating order, requiring no LLM involvement.

3.3 EduIllustrate Benchmark

Dataset Construction. We construct a curated benchmark derived from the K12-Vista dataset (Zhang et al., 2025), a large-scale repository of Chinese K-12 STEM problems with gold-standard solutions and fine-grained knowledge point annotations. We extract 118 problems following three curation criteria: **(1) Diagram-Appropriate Problems**—problems where visual representations convey spatial, topological, or relational information essential to problem-solving, rather than merely restating text (Ji et al., 2025); **(2) Clear Gold Solutions**—verified step-by-step solutions with intermediate reasoning steps, enabling objective correctness evaluation; **(3) Diverse Knowledge Points**—sampling to maximize diversity at the 3rd level of K12-Vista's 4-level knowledge point taxonomy, ensuring varied conceptual coverage. The full knowledge point breakdown is provided in Appendix A.

Grade	Math	Phys	Chem	Bio	Geo	Total
Elementary	10	—	—	—	—	10
Middle	20	20	3	5	5	53
High	20	20	5	5	5	55
Total	**50**	**40**	**8**	**10**	**10**	**118**

Table 1: Benchmark distribution across grade levels and subjects.

The benchmark comprises 118 problems across three grade levels (elementary, middle, high school) and five STEM subjects, as shown in Table 1. All 118 problems include original diagrams in the problem statement, requiring models to interpret visual input and generate explanatory diagrams that reference or extend the given figure. Problems require 2–6 solution steps (mean = 3.8, SD = 1.2), and collectively span 102 distinct knowledge points at the 3rd taxonomy level. Figure 3 shows the knowledge point categories by subject.

Evaluation Framework. We propose an **8-dimension rubric** grounded in multimedia learning theory (Sweller, 1988; Paivio, 1971; Mayer, 2001), covering three aspects of multimodal educational content. Each dimension is scored on a 0–5 Likert scale.

Text quality (evaluated on extracted <TEXT_k> blocks): Correctness & Completeness—whether the explanation reaches the correct answer through valid reasoning with all intermediate steps; Logical Coherence—whether reasoning flows naturally from premises to conclusions; Pedagogical Effectiveness—whether explanations use grade-appropriate language and effective instructional

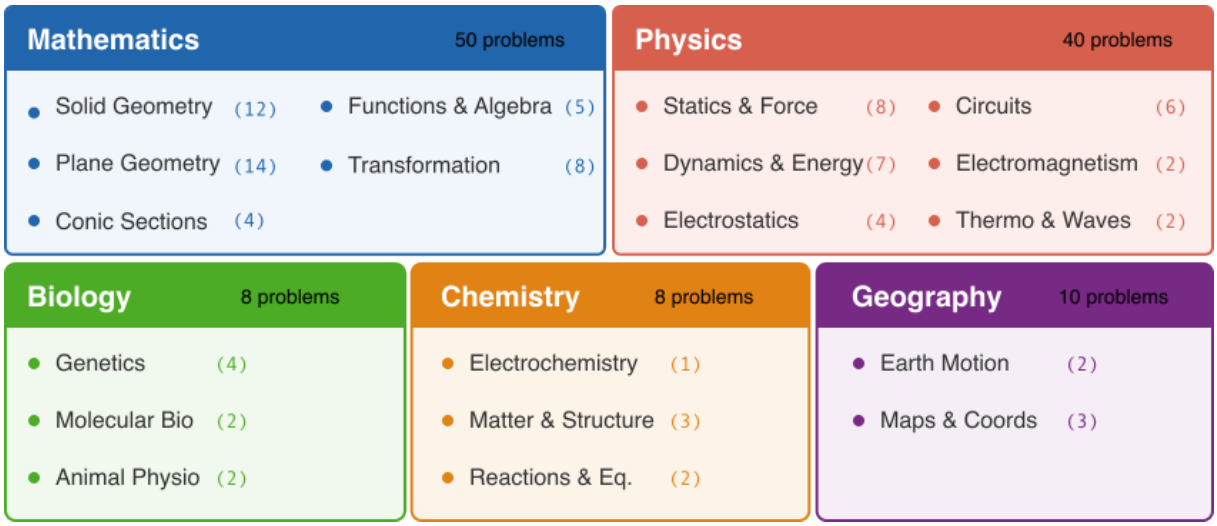

Figure 3: Knowledge point categories by subject. Values in parentheses indicate the number of unique knowledge points mapped in each category.

strategies; Typographic Clarity—proper mathematical notation, consistent formatting, and absence of artifacts.

Visual quality (evaluated on rendered diagrams): Diagram–Problem Alignment—whether diagrams faithfully represent the problem's geometric or physical setup; Element Layout Quality—whether elements avoid overlap, maintain readable spacing, and follow discipline conventions; Visual Consistency—whether multiple diagrams maintain coherent color schemes, labeling, and style; Text–Diagram Coordination—how well textual references integrate with the corresponding diagrams.

Automated Evaluation Protocol. We employ Gemini 3.0 Pro Preview (temperature=0) as the judge model, selected for its strong multimodal reasoning and 2M-token context window. Evaluation input varies by dimension type: text-only dimensions (Correctness & Completeness, Logical Coherence, Pedagogical Effectiveness, Typographic Clarity) receive the problem statement, extracted text, and rubric criteria (plus the gold solution for Correctness & Completeness); multimodal dimensions (Diagram–Problem Alignment, Text–Diagram Coordination) additionally receive the relevant rendered diagrams alongside surrounding text. For Element Layout Quality, each diagram is scored independently and aggregated via geometric mean $ELQ = \left(\prod_{i=1}^{N} s_i\right)^{1/N}$, ensuring a single poor diagram substantially penalizes the overall score. For Visual Consistency, Scene 1 serves as the visual anchor; each subsequent diagram is compared against it, with $VC = \left(\prod_{i=2}^{N} s_{6,i}\right)^{1/(N-1)}$.

Human Evaluation Protocol. To validate LLM-as-judge reliability, we conduct human evaluation on 30 stratified random sampled explanations, scored by 20 expert raters across 7 dimensions (Correctness & Completeness excluded) on a 3-level scale {0, 0.5, 1}, producing 4,200 judgments. To enable direct comparison with the automated 0–5 Likert scores, LLM scores are normalized to [0, 1] by dividing by 5 before computing Spearman's ρ. We compute Krippendorff's α for inter-rater agreement and Spearman's ρ for human-AI correlation. Full annotation procedures are described in Appendix E.

4 Experimental Results

We evaluate five frontier LLMs on the EduIllustrate benchmark, analyzing performance across dimensions, subjects, and grade levels. We validate our sequential workflow design through ablation studies and assess LLM-as-judge reliability via human evaluation with 20 expert raters.

4.1 Experimental Setup

Models Evaluated. We test five state-of-the-art LLMs: (1) Gemini 3.0 Pro Preview (Google DeepMind), (2) GPT-5 (OpenAI), (3) Claude Sonnet 4.5 (Anthropic), (4) Kimi-K2.5 (Moonshot AI), and (5) Qwen 3.5-35B (Alibaba Cloud). All models use identical prompts with temperature=0.7 on all 118 benchmark problems.

Model	C&C	LC	DPA	PE	ELQ	VC	TDC	TC	Overall
Gemini 3.0 Pro Preview	4.45	4.67	4.44	3.91	4.95	4.23	4.42	4.67	**4.40**
Kimi-K2.5	4.50	4.70	3.51	3.78	4.85	3.35	4.48	4.20	**4.07**
Claude Sonnet 4.5	3.09	3.17	2.46	2.69	3.91	3.24	4.33	3.64	**3.12**
GPT-5	3.57	4.03	2.44	2.54	3.27	3.00	4.57	3.51	**3.11**
Qwen 3.5-35B	4.29	4.58	1.80	3.50	4.30	2.50	4.13	2.46	**2.98**

Table 2: Model performance across 8 evaluation dimensions (0–5 scale, 118 problems). C&C: Correctness & Completeness, LC: Logical Coherence, DPA: Diagram–Problem Alignment, PE: Pedagogical Effectiveness, ELQ: Element Layout Quality, VC: Visual Consistency, TDC: Text–Diagram Coordination, TC: Typographic Clarity. Overall score computed as geometric mean.

4.2 Main Results

Table 2 presents overall performance across all eight dimensions. The overall score is computed as the geometric mean of all eight dimension scores, ensuring that a single critically low score substantially penalizes the overall result. Gemini 3.0 Pro Preview achieves the highest overall score (4.40/5.0), substantially outperforming all other models. Kimi-K2.5 ranks second (4.07), while Claude Sonnet 4.5, GPT-5, and Qwen 3.5-35B cluster at 2.98-3.12. The 32% performance gap between best and worst models demonstrates that architectural choices significantly impact multimodal educational content quality.

All models excel at Logical Coherence and Typographic Clarity (scores $\geq$ 3.17), reflecting LLMs' strength in coherent, well-formatted text generation. Text–Diagram Coordination achieves high scores even for weaker models (Claude: 4.33, GPT-5: 4.57), indicating models produce well-integrated textual references to diagrams regardless of diagram quality. Critical weaknesses appear in Diagram–Problem Alignment, where scores range from 1.80 to 4.44. Qwen's catastrophic score of 1.80 reflects frequent semantic misalignment—geometrically valid diagrams that do not match problem specifications. Pedagogical Effectiveness is universally weakest (2.54-3.91), indicating all models struggle with grade-appropriate language and scaffolding strategies. Qwen encountered 11 complete failures (9.3% failure rate), primarily on high school physics and chemistry problems, while GPT-5 had 2 failures (1.7%). All other models completed all 118 problems.

Table 3 shows subject- and grade-level performance. Mathematics and physics consistently achieve highest scores, while geography scores lowest (2.25–4.07). Gemini and Kimi maintain consistency across grade levels (Gemini: 4.27–4.42, Kimi: 3.97–4.13), while other models degrade substantially on high school problems. GPT-5's sharp drop from elementary (3.85) to high school (3.00) and Qwen's 9.3% failure rate concentrated on high school problems both correlate with increasing diagram complexity at higher grade levels.

4.3 Workflow Ablation Study

EduIllustrate is built upon a modified version of the TheoremExplainAgent codebase (Ku et al., 2025), which originally adopts an all-parallel workflow where all scenes generate implementation plans and code in parallel. We compare our sequential Scene 1 + parallel Scenes 2-N workflow against this all-parallel baseline to quantify the impact of our anchoring strategy. The baseline generates and renders all scenes in parallel without Scene 1 conditioning. Table 4 shows the all-parallel approach incurs 32% higher input token consumption (100.3K vs. 75.8K)—because every scene independently generates a full implementation plan before code generation, whereas our workflow restricts planning to Scene 1 only—17.5% higher cost ($1.14 vs. $0.97), 48% longer latency (7.32 vs. 4.94 minutes)—attributable to the additional model calls required for per-scene planning—, and 12% worse Visual Consistency (3.89 vs. 4.42), while achieving lower overall quality (4.29 vs. 4.40). Manual inspection reveals the all-parallel workflow's consistency failures stem from independent scene generation preventing style propagation. A side-by-side visual comparison is provided in Appendix C (Figure 6). This validates our sequential anchoring strategy.

4.4 Human-AI Agreement Analysis

Table 5 presents inter-rater reliability (Krippendorff's α) and human-AI agreement (Spearman's ρ) for 30 explanations evaluated across 7 dimensions by 20 expert raters. Logical Coherence and Diagram–Problem Alignment achieve strong human consensus ($\alpha \geq 0.80$) and strong human-

Model	Math	Phys	Chem	Bio	Geo	Elem.	Middle	High
Gemini 3.0 Pro Preview	4.42	4.44	4.48	4.43	4.07	4.27	4.41	4.42
Kimi-K2.5	4.17	4.02	4.13	4.17	3.71	3.97	4.03	4.13
Claude Sonnet 4.5	3.17	3.15	3.37	3.06	2.65	3.32	3.04	3.17
GPT-5	3.16	3.05	3.81	3.03	2.55	3.85	3.09	3.00
Qwen 3.5-35B	3.19	2.83	3.18	3.05	2.25	3.64	2.93	2.90

Table 3: Model performance by subject (left) and grade level (right), overall score on 0–5 scale. Due to small sample sizes (Chem: n=8, Bio: n=10, Geo: n=10), cross-subject comparisons for these disciplines should be treated as exploratory and do not support causal inference.

Workflow	Tokens (In/Out)	Cost ($)	Latency (min)	VC	Overall
Ours	75.8K / 68.1K	0.97	4.94	4.42	**4.40**
All-Parallel	100.3K / 78.5K	1.14	7.32	3.89	**4.29**

Table 4: Workflow comparison using Gemini 3.0 Pro Preview on 118 problems. Our workflow achieves superior visual consistency at lower cost and latency.

Dim.	Spearman ρ	Krippendorff α	Reliability
LC	0.89	0.84	Strong
DPA	0.83	0.80	Strong
TC	0.77	0.67	Strong-Mod.
PE	0.70	0.68	Moderate
TDC	0.66	0.64	Moderate
VC	0.47	0.30	Weak
ELQ	0.39	0.32	Weak

Table 5: Human-AI agreement and inter-rater reliability (30 explanations, 20 raters). Correctness & Completeness excluded (objective gold answer verification).

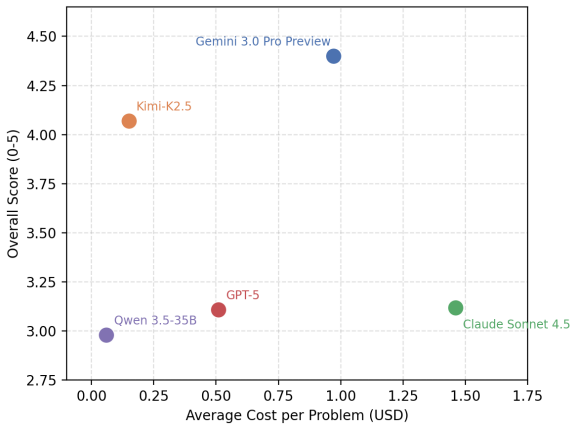

Figure 4: Cost vs. quality trade-off across five models. X-axis: average cost per problem (USD); Y-axis: overall score (0–5). Kimi-K2.5 offers the best cost-efficiency, achieving 4.07 at $0.15/problem.

AI agreement ($\rho \geq 0.80$), validating LLM-as-judge for these dimensions. Logical Coherence's high reliability reflects that logical gaps are objectively identifiable; Diagram–Problem Alignment's strong performance is notable given it requires visual reasoning. Typographic Clarity achieves strong-to-moderate agreement ($\rho = 0.77$, $\alpha = 0.67$), as formatting artifacts are largely objective but occasionally ambiguous at borderline cases. Pedagogical Effectiveness and Text–Diagram Coordination show moderate reliability ($\alpha \approx 0.64 - 0.68$, $\rho \approx 0.66 - 0.70$), acceptable for comparative studies but requiring human validation for high-stakes decisions. Visual Consistency and Element Layout Quality exhibit weak reliability ($\alpha \approx 0.30$, $\rho \approx 0.40$). Beyond ill-defined rubric constructs, a key contributing factor is that the LLM judge model is insensitive to low-level visual artifacts—such as overlapping elements, misaligned labels, or rendering glitches—that human raters readily detect when inspecting rendered diagrams.

4.5 Cost Analysis

Table 6 presents cost and latency metrics. As shown in Figure 4, Kimi offers the best cost-efficiency at $0.15/problem (4.07 score), representing only 7.5% quality degradation relative to Gemini ($0.97, 4.40 score) at 10× lower cost. Qwen is cheapest ($0.06) but sacrifices 32% quality.

5 Conclusion

We presented EduIllustrate, a four-stage agentic pipeline with sequential Scene 1 anchoring, a 118-problem benchmark, and an 8-dimension rubric grounded in multimedia learning theory. Sequential anchoring improves Visual Consistency by 12% over fully parallel generation at 17.5% lower cost. Human evaluation validates LLM-as-judge reliability on objective dimensions while revealing

Model	Avg Input Tokens	Avg Output Tokens	Input ($/M)	Output ($/M)	Avg Cost ($)	Latency (min)
Claude Sonnet 4.5	90,439	79,035	3.00	15.00	1.46	6.92
Gemini 3.0 Pro Preview	75,802	68,093	2.00	12.00	0.97	4.94
GPT-5	66,420	42,677	1.25	10.00	0.51	7.67
Kimi-K2.5	58,689	54,893	0.45	2.20	0.15	6.34
Qwen 3.5-35B	61,883	37,737	0.16	1.30	0.06	5.34

Table 6: Cost and latency analysis per problem (118 problems, average values).

limitations on subjective visual assessment.

Limitations

Static Diagrams Only. The pipeline currently generates static PNG images, though the agent framework can be straightforwardly adapted to produce process animations by removing the `-s` flag in Manim rendering and adjusting the prompts. The primary limitation is the absence of a systematic evaluation protocol for animated outputs—our 8-dimension rubric does not cover temporal coherence or animation pacing, which we leave for future work.

LLM-as-Judge Visual Assessment Limitations. Weak human-AI agreement on visual dimensions (Element Layout Quality: $\rho = 0.39$, Visual Consistency: $\rho = 0.47$) indicates current vision-language models struggle to assess layout quality and stylistic consistency reliably. Low inter-rater agreement among humans ($\alpha \approx 0.30$) suggests these constructs are ill-defined, requiring rubric refinement through exemplar diagrams for each score level or decomposition into objective binary checks (e.g., element overlap detection, color scheme extraction and comparison). Future work should explore specialized visual quality assessment models fine-tuned on human-annotated diagram datasets.

Technical Implementation Constraints. The pipeline is tightly coupled to Manim, limiting generalizability to other frameworks (TikZ, matplotlib, SVG). Extending to target-agnostic intermediate representations as proposed by Kumar et al. (2025) would improve flexibility.

Pedagogical Effectiveness. Pedagogical Effectiveness is the weakest dimension across all models (scores 2.54–3.91), reflecting a fundamental limitation: the current pipeline delivers static, one-shot explanations with no adaptation to the individual learner. The system has no mechanism for grade-appropriate scaffolding, Socratic questioning, or student-specific personalization. Future work will pursue two directions to address this. First, incorporating instructional scaffolding to enable multi-turn interaction with students—guiding them through problems via Socratic dialogue rather than delivering static explanations. Second, integrating student profiles (personality traits, cognitive styles, interests) to personalize both the explanation strategy and diagram design for individual learners.

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A Benchmark Construction Details

Dataset Curation Process

Benchmark construction followed a two-phase procedure. In the first phase, Kimi-K2.5 automatically screened all candidate problems from K12-Vista (Zhang et al., 2025) for **Diagram Appropriateness**: for each problem, the model was prompted to judge whether a visual representation would convey spatial, topological, or relational information essential to problem-solving (rather than merely restating the text), and to output a binary decision with a brief justification. In the second phase, two human annotators independently reviewed every problem against all three curation criteria—**(1) Diagram-Appropriate Problems**, **(2) Clear Gold Solutions**, and **(3) Diverse Knowledge Points**—and resolved disagreements through discussion, using a dedicated review interface (Figure 5). Problems were retained only when both annotators confirmed all three criteria were satisfied. This two-phase design combines the scalability of LLM screening with the reliability of human expert judgment.

Knowledge Point Coverage

Table 7 lists the complete knowledge point coverage of the 118 benchmark problems, organized by grade level and subject.

B Judge Model Self-Preference Analysis

A potential concern with using Gemini 3.0 Pro Preview as the judge model is self-preference bias: the judge may systematically inflate scores for outputs it generated. To investigate this, we sampled 50 problems (the same problems across all models) and re-evaluated all outputs using GPT-5 as an alternative judge, then compared the two scoring distributions.

Table 8 reports overall scores assigned by each judge. Two observations address the self-preference concern. First, both judges agree on the **ranking** of all five models—Gemini $>$ Kimi $>$ Claude $\approx$ GPT-5 $>$ Qwen—suggesting the relative conclusions of our main results are robust to judge choice. Second, GPT-5 assigns *higher* absolute scores to Gemini (4.76 vs. 4.40) and to most other models, indicating that if any self-preference exists in Gemini-as-judge, it does not inflate Gemini's relative standing. The score gap between Gemini and the next-best model (Kimi) is 0.33 under Gemini-as-judge and 0.57 under GPT-5-as-judge, further confirming that Gemini's top position is not an artifact of self-scoring. We therefore conclude that the main findings are not materially affected by judge model self-preference.

C Visual Consistency Case Study

To illustrate the impact of our sequential Scene 1 anchoring strategy on Visual Consistency, we present a side-by-side comparison of two explanations generated for the same high school mathematics problem (four-pyramid P-$ABCD$, finding $NC : NP$ given $MN \parallel$ plane PAD). Both expla-

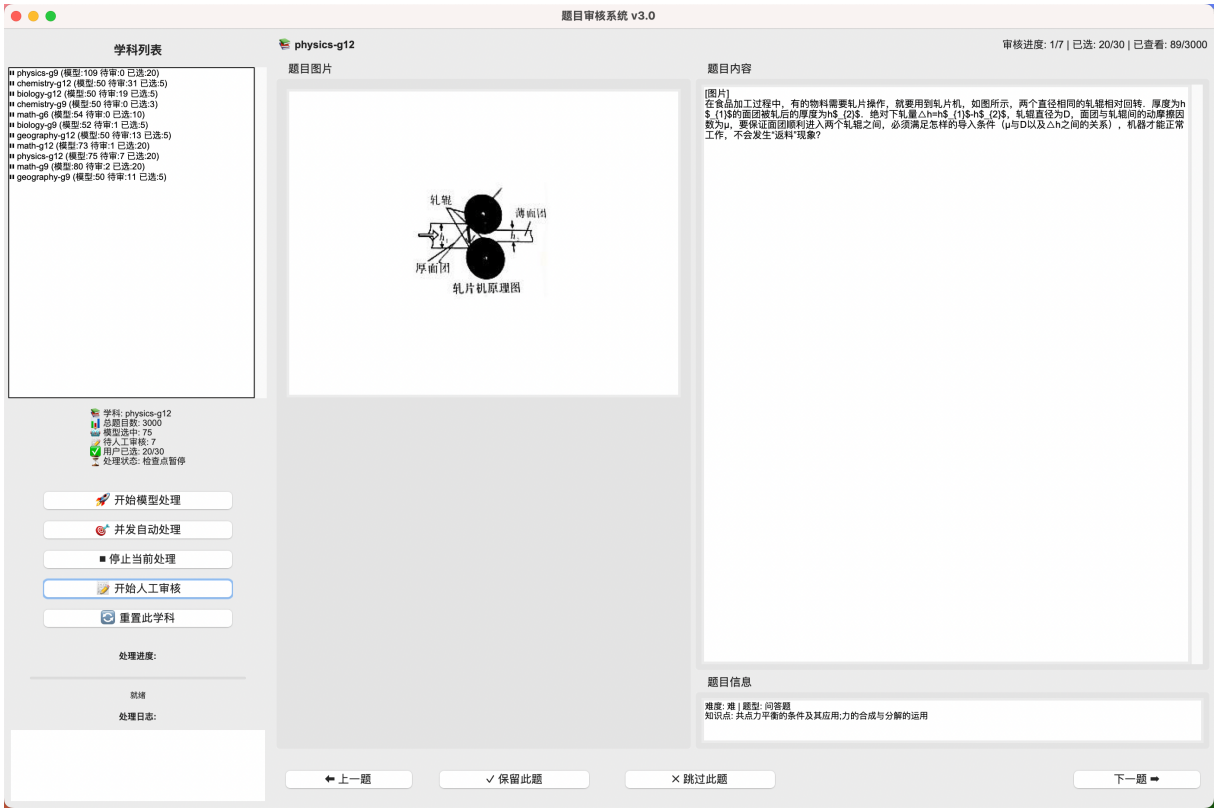


Figure 5: Screenshot of the problem review interface used during human annotation. Annotators evaluated each candidate problem against the three curation criteria and recorded their decisions with justifications.

nations are generated by Gemini 3.0 Pro Preview: the top row uses our workflow (sequential Scene 1 planning + parallel Scenes 2–N conditioned on Scene 1's code), while the bottom row uses the all-parallel baseline (all scenes planned and generated independently).

D A Gallery of Generated Explanations

We present representative high-quality and low-quality explanations generated by EduIllustrate to illustrate the range of outputs across different subjects and failure modes. Each entry shows the problem statement, key solution steps, and the corresponding rendered diagrams.

High-Quality Explanations

Example 1: Middle School Physics (Circuit Analysis)

Example 2: Middle School Biology (DNA Structure & Replication)

Example 3: High School Math (Infinite Geometric Series, Inscribed Circles)

Example 4: High School Math (Solid Geometry, Perpendicular Planes)

Example 5: Middle School Math (Quadratic Function Coefficients)

Example 6: Middle School Math (Square and Rhombus, Shaded Area)

Low-Quality Explanations

Example 7: Elementary Math (Cube Net Folding) — Wrong Final Answer

Example 8: Middle School Math (Isosceles Triangle) — Diagram Inconsistent with Problem

Example 9: High School Physics (Gas Laws, Two Cylinders)— Flawed Diagram

Example 10: Middle School Geography (Latitude/Longitude Reading) — Flawed Diagram Reading and Reasoning

E Human Annotation Process

Human evaluation was conducted through a dedicated annotation website. Raters were presented with a K-12 STEM problem, its gold-standard solution, and the generated illustrated explanation (including rendered diagrams). For each explanation, raters scored 7 dimensions (Logical Coherence through Typographic Clarity) on a coarse 3-level ordinal scale {0, 0.5, 1}, where 0 indicates

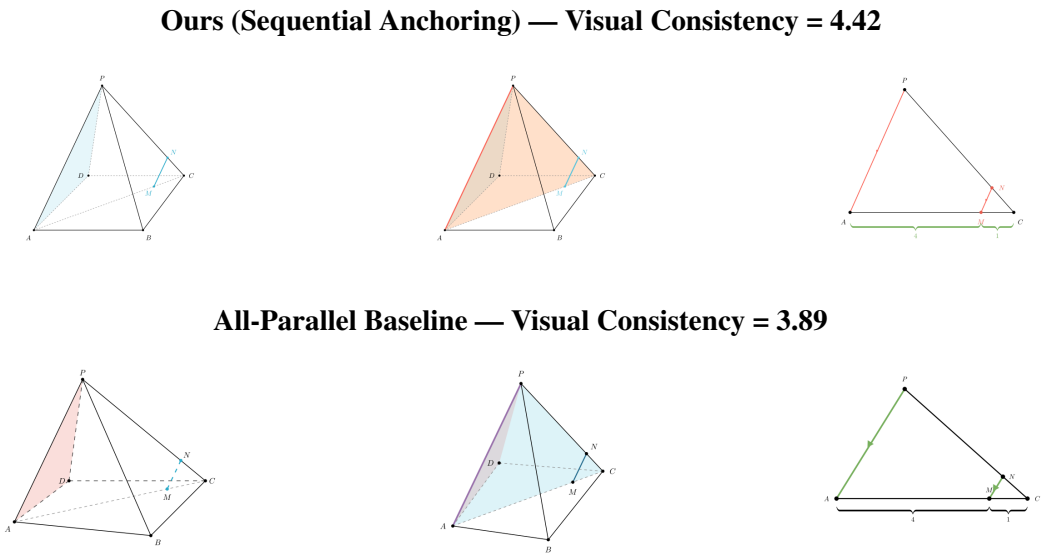


Figure 6: Visual consistency comparison between our sequential anchoring workflow (top) and the all-parallel baseline (bottom) on the same problem (four-pyramid P-$ABCD$). In our workflow, Scene 1 and Scene 2 render the identical pyramid structure—Scene 2 simply highlights the PAC face on top of the same base diagram—maintaining full geometric and stylistic consistency. In the all-parallel baseline, the pyramid in Scene 1 and Scene 2 is drawn differently, breaking visual coherence and forcing the reader to reconcile two inconsistent representations of the same object.

poor quality, 0.5 indicates acceptable quality, and 1 indicates high quality. Correctness & Completeness was excluded from human evaluation because solution correctness can be determined objectively by comparing against the gold-standard answer, making subjective human judgment unnecessary and potentially inconsistent. Each page displayed one explanation at a time, with the rubric description and score criteria shown alongside. Raters could zoom into individual diagrams before scoring visual dimensions.

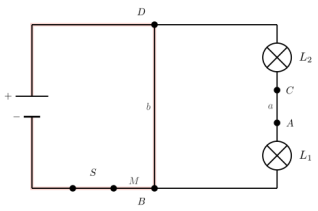
Rater Training. Before formal scoring, all 20 raters underwent a calibration session. For each of the 7 dimensions, we presented two anchor examples—one high-quality explanation demonstrating exemplary visual and textual clarity, and one low-quality explanation exhibiting common failure modes (e.g., overlapping diagram elements, misaligned labels, or pedagogically ineffective step sequencing). Particular emphasis was placed on the aesthetic dimensions (Element Layout Quality and Visual Consistency), as these proved most subjective. Raters discussed borderline cases together to align their interpretation of the 3-level scale.

Pilot Scoring. Following training, raters completed a pilot round on 5 held-out explanations not included in the final evaluation set. Inter-rater agreement was computed via Krippendorff's α after the pilot. Raters whose individual agreement with the group fell below an acceptable threshold were given additional feedback before proceeding. The pilot Krippendorff's α across all dimensions exceeded 0.60, confirming sufficient calibration to proceed with formal scoring.

Formal Scoring. Each of the 30 explanations in the final evaluation set was independently scored by all 20 raters, producing $30 \times 7 \times 20 = 4{,}200$ human judgments. Figure 7 shows a screenshot of the annotation interface.

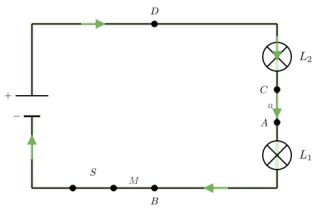
Problem: The circuit is shown. Which analysis is incorrect? (A) After S is closed, the circuit short-circuits. (B) After S is closed, L_1 and L_2 are in parallel and both light up. (C) Removing wire b makes L_1 and L_2 series-connected. (D) Moving wire M from B to A makes L_1 and L_2 parallel.

Approach: Trace current paths from the positive terminal to determine circuit topology under each scenario, then evaluate each option.



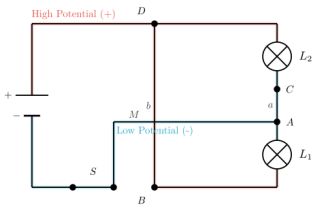
Scene 1: Original circuit diagram with current path analysis. Wire b short-circuits the bulbs, creating a zero-resistance path.

Step 2: Removing wire b forces current through $L_2 \rightarrow L_1$ sequentially—series connection confirmed.



Scene 2: Modified circuit with wire b removed. Single current path through both bulbs in series.

Step 3: Moving wire M from B to A creates two independent paths. Both bulbs connect between the same high/low potential nodes—parallel connection.



Scene 3: Circuit with wire M moved to A. Both bulbs span identical potential difference—parallel.

Answer: B (bulbs do not light up; short circuit prevents current through loads).

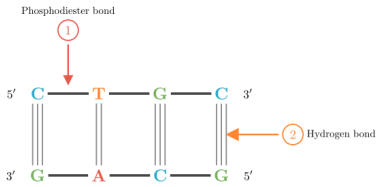
High-quality example 1. Diagrams consistently use the same circuit drawing conventions across all three scenes. Each scene incrementally modifies the previous one, supporting step-by-step reasoning.

Level	Subject	Knowledge Points
High School	Biology (5)	Law of independent assortment; PCR amplification of DNA; Variants of 9:3:3:1 and 1:1:1:1 ratios; Sex-linked inheritance; Determination of gene locus on chromosomes
	Chemistry (5)	Galvanic and electrolytic cell principles (Œ2); Crystal structure and properties; Chemical equilibrium (equivalent equilibrium)
	Geography (5)	Earth and maps; General characteristics of Earth's motion
	Mathematics (20)	Proportional segments related to circles; Hyperbola properties; Proposition truth and plane-plane relations; Proposition truth and pyramid structure; Inscribed polygon properties; Tangent line proof and proportional segments; Common features of conic sections; Plane-plane perpendicularity; Parabola properties; Limits of sequences; Ellipse properties; Law of sines; Sphere volume and surface area; Area/volume from three-view drawings; Spatial line-line relations; Arithmetic sequences; Spatial vectors and sphere; (some topics unlabeled)
	Physics (20)	Concurrent force equilibrium and electric field strength; Work-energy theorem and projectile motion; Work-energy theorem and charged particle in uniform electric field; Work-energy theorem and mechanical energy conservation; Conservation of momentum; Momentum, mechanical energy, and energy conservation; EMF from conductor motion and Joule's law (Œ3, various combinations); Charged particle in uniform electric field (Œ2, with work-energy theorem); Coulomb's law and Newton's second law; Mechanical energy conservation and projectile; Ideal gas law and enclosed gas pressure; SHM amplitude/period/frequency, vertical projection, Hooke's law; Energy conservation, steady current, and force equilibrium
Middle School	Biology (5)	DNA replication/transcription/translation calculations; Complete and incomplete metamorphosis in insects; Urinary system and urine formation
	Chemistry (3)	Molecules, atoms, ions, elements and matter; Molecular definition and acid-base indicators; Chemical inference, decomposition and displacement reactions
	Geography (5)	Latitude and longitude; Principles of day length variation and seasons; Determining direction and location using coordinate grids (Œ3)
	Mathematics (20)	Linear function applications; Triangle angle sum and exterior angles; Triangle angle sum and rotation; Circumscribed circle and circumcenter; Quadratic/linear function graphs and coefficients; Triangle congruence (SAS) and Pythagorean theorem; 3D solid nets; Similar triangles (judgment and properties); Geometric solid nets; Inequalities, linear and quadratic functions (comprehensive); 30° right triangle, inscribed angles, and diameter angles; Net folding; Proportional segments, midsegment, and 30° triangle; Rotation and area; Square properties and coordinates; Similar triangles and equilateral triangle; Fold transformations; Axial symmetry; Golden ratio and right triangle solving
	Physics (20)	Force balance and pulley systems; Diagnosing circuit faults with ammeter/voltmeter; Force and motion; Pressure comparison, density, gravity vs. pressure; Balanced forces and kinetic/potential energy; Spring force meter; Friction force; Lever equilibrium (Œ2, including minimum force); Ohm's law and electromagnet factors; Ohm's law applications; Pulley rope tension, work, and power; Ammeter usage; Three circuit states; Dynamic circuit analysis (Œ2, with power); Magnetic pole interactions and Ampere's right-hand rule; Archimedes' principle and force balance (Œ2)
Elementary	Mathematics (10)	Three-view and net diagrams; Observing 3D shapes from different views; Shape composition; Position and direction; Rotation and coordinate position; Perimeter of composite plane figures; Overlapping problems; Cube/cuboid nets and folding

Table 7: Complete knowledge point coverage of the 118 EduIllustrate benchmark problems. Numbers in parentheses indicate problem count per subject-level combination. Some high school mathematics problems have unlabeled knowledge points in the original K12-Vista dataset.

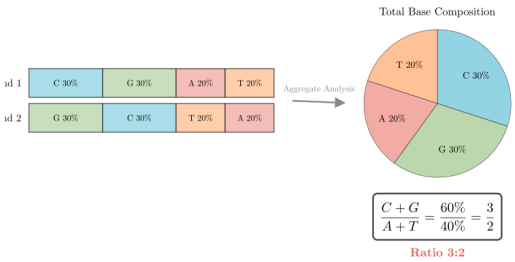
Problem: A ^{15}N-labeled eukaryotic gene has Cytosine (C) accounting for 30% of all bases. Which statement is correct? (A) Helicase acts on sites ① and ②. (B) In one strand, (C+G)/(A+T) = 3:2. (C) After $T \rightarrow A$ mutation at site ①, after n replications the mutant gene is 1/4. (D) After 3 replications in ^{14}N medium, DNA with ^{14}N is 3/4.

Approach: Identify bond types at sites ① and ②; apply Chargaff's rules; apply semi-conservative replication.



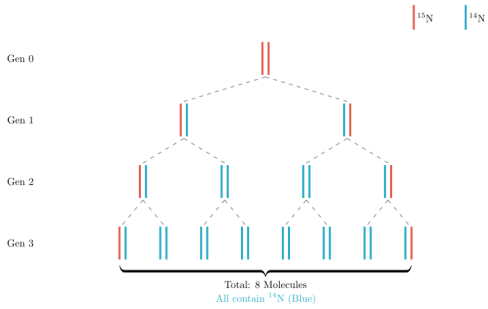
Scene 1: DNA double helix with sites ① (hydrogen bonds between bases) and ② (phosphodiester bonds in backbone) clearly labeled.

Step 2: Chargaff's rules: C=G=30%, so A=T=20%. Within one strand, C+G=60%, A+T=40%, giving ratio 3:2—Statement B is correct.



Scene 2: Base composition diagram showing C/G/A/T proportions derived from Chargaff's rules.

Step 3: Semi-conservative replication after 3 rounds yields $2^3 = 8$ DNA molecules; all 8 contain ^{14}N (since new strands use ^{14}N); 2 of 8 also retain ^{15}N. So ^{14}N proportion = 8/8 = 100%, not 3/4—Statement D is wrong.



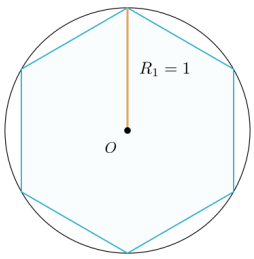
Scene 3: Semi-conservative replication tree after 3 rounds, showing isotope distribution across 8 daughter molecules.

Answer: B.

High-quality example 2. Diagrams use consistent color coding (blue for ^{15}N strands, orange for ^{14}N strands) across all scenes, with each scene building directly on the previous one.

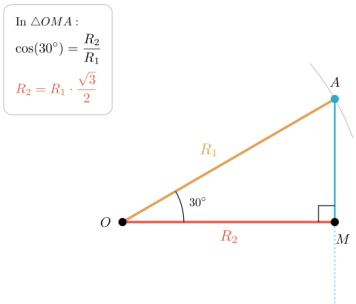
Problem: A regular hexagon is inscribed in a circle of radius 1 m. Its inscribed circle is drawn, inside which another regular hexagon is inscribed, and so on infinitely. Find the sum S of the areas of all circles.

Approach: Find the ratio between successive circle radii using the apothem of a regular hexagon, identify the geometric series common ratio, then apply the infinite sum formula.



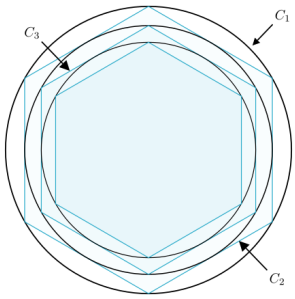
Scene 1: First circle ($R_1 = 1$) with inscribed hexagon and its apothem shown. The apothem equals $\frac{\sqrt{3}}{2} R_1$, giving $R_2 = \frac{\sqrt{3}}{2}$.

Step 2: Area ratio $= \left(\frac{R_2}{R_1}\right)^2 = \frac{3}{4}$. The areas form a geometric series with first term π and common ratio $\frac{3}{4}$.



Scene 2: First two circles and hexagons overlaid, illustrating the scaling relationship.

Step 3: $S = \frac{\pi}{1-\frac{3}{4}} = 4\pi \text{ m}^2$.



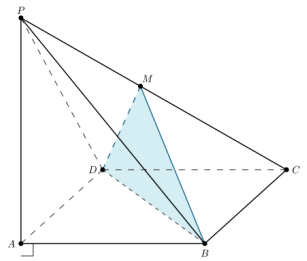
Scene 3: Infinite nesting visualized with converging circles and hexagons.

Answer: $S = 4\pi$ **m**^{2}**.**

High-quality example 3. Each scene builds progressively on the previous geometric construction, with consistent styling and annotations.

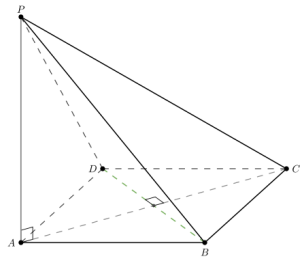
Problem: In quadrangular pyramid P-$ABCD$, $PA \perp$ base $ABCD$, all base edges are equal (rhombus). M is a moving point on PC. State a condition on M such that plane $MBD \perp$ plane PCD.

Approach: Use the plane-perpendicularity theorem: find a line in plane MBD perpendicular to plane PCD. Show $BD \perp PC$ using the rhombus diagonal and $PA \perp$ base properties, then determine where MBD must intersect PCD.



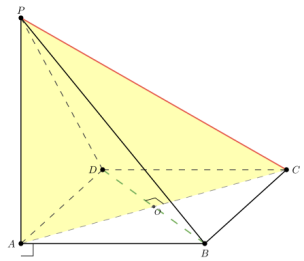
Scene 1: Pyramid with rhombus base $ABCD$, PA perpendicular to base, diagonal BD highlighted.

Step 2: Since $ABCD$ is a rhombus, $AC \perp BD$. Since $PA \perp$ base, $PA \perp BD$. Thus $BD \perp$ plane PAC, hence $BD \perp PC$.



Scene 2: Plane PAC highlighted, showing $BD \perp$ plane PAC and therefore $BD \perp PC$.

Step 3: $BD \perp PC$ and $BD \subset$ plane MBD; if M is the foot of the altitude from B to PC (i.e., $MB \perp PC$), then $BD \perp$ plane PCD, giving plane $MBD \perp$ plane PCD.



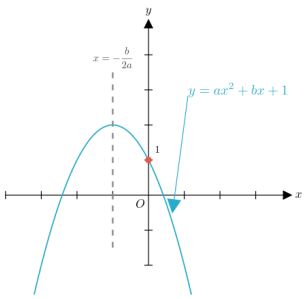
Scene 3: Plane MBD and plane PCD shown with their intersection line, confirming perpendicularity condition.

Answer: $MB \perp PC$ **(i.e.,** M **is the foot of the perpendicular from** B **to** PC**).**

High-quality example 4. Three scenes incrementally reveal the geometric reasoning: base properties, perpendicularity proof, and final plane configuration.

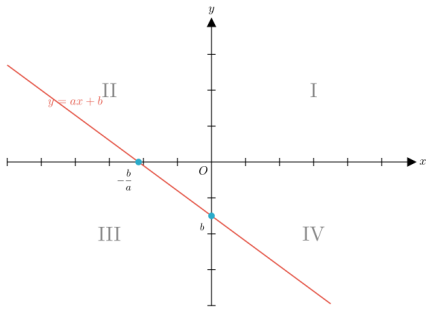
Problem: The graph of $y = ax^2 + bx + 1$ is shown. Which quadrant does $y = ax + b$ *not* pass through?
A. First B. Second C. Third D. Fourth

Approach: Read signs of a and b from the parabola (opening direction, axis of symmetry position), then determine the linear function's slope and intercept to identify the missing quadrant.



Scene 1: Parabola opens downward ($a < 0$), axis of symmetry at $x < 0$, so $-b/(2a) < 0 \Rightarrow b < 0$.

Step 2: For $y = ax + b$: slope $a < 0$ (decreasing), y-intercept $b < 0$. A decreasing line with negative intercept passes through Quadrants II, III, IV—it does *not* pass through Quadrant I.



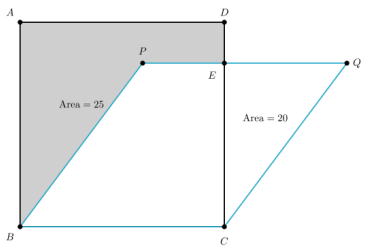
Scene 2: Linear function $y = ax + b$ plotted with $a < 0$, $b < 0$, passing through Q2, Q3, Q4 only.

Answer: A (First quadrant).

High-quality example 5. The parabola and linear function are rendered with consistent axis styles and color coding, with the missing quadrant clearly shaded.

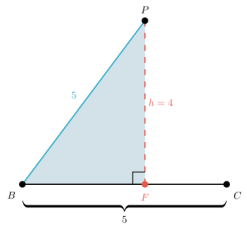
Problem: Square $ABCD$ has area 25; rhombus $PQCB$ has area 20. Find the area of the shaded region.
A. 11 B. 6.5 C. 7 D. 7.5

Approach: Determine side lengths from given areas; use the subtraction method to express the shaded area as the square minus an unshaded triangular region formed by the overlap of the square and rhombus.

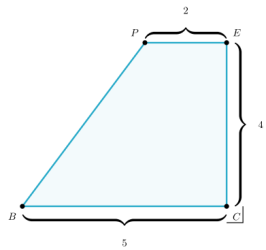


Scene 1: Square $ABCD$ (side=5) and rhombus $PQCB$ overlaid; the shaded region and the unshaded overlap region clearly distinguished by color.

Step 2: Side of square = 5. Height of rhombus = area/base = 20/5 = 4. The unshaded region inside the square is a right triangle with legs 5 and $5-4=1$; its area = $\frac{1}{2}\times 5\times 1 = 2.5$. Shaded area $= 25 - 2\times 2.5\times \ldots$ (subtraction applied per configuration).



Scene 2: Decomposition of the unshaded region into computable triangles.



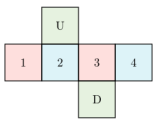
Scene 3: Final shaded region highlighted with computed area.

Answer: D (7.5).

High-quality example 6. Consistent color scheme across scenes: the square in blue, rhombus in orange, shaded region in green. Each scene progressively focuses on the region of interest.

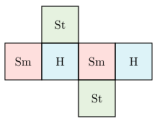
Problem: Xiaoxiao made a cubic gift box where opposite faces share the same pattern ($\star$, $\heartsuit$, $\odot$). Which net (A/B/C/D) folds into this cube?

Approach: Apply opposite-face pairing rule for cross-type nets: positions (1,3), (2,4), and (top, bottom) are opposite pairs.



Scene 1: Cross-type net with positions labeled and opposite-pair rule illustrated.

Step 2: Tests each option by mapping symbols to opposite-pair positions. Concludes that option C satisfies the constraint.



Scene 2: Net options annotated with opposite-face verification.

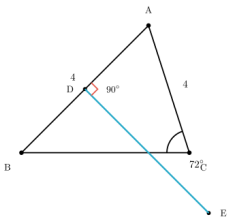
Final Answer: C $\times$ **Gold answer: A**

Failure mode (Correctness & Completeness): The reasoning process applies the opposite-face pairing rule correctly in principle, but misidentifies the symbol placement in option A vs. C during the mapping step, leading to an incorrect final answer.

Low-quality example 7 (Correctness & Completeness failure). The model's final answer (C) contradicts the gold standard answer (A). Despite a structured approach, an error in reading the net topology causes the incorrect conclusion.

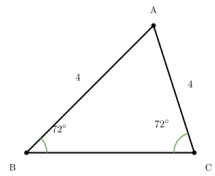
Problem: In $\triangle ABC$, $AB = AC = 4$, $\angle C = 72^\circ$. Point D is the midpoint of AB; point E lies on AC; $DE \perp AB$. Find $\cos A$.
Options: A. $\frac{\sqrt{5}-1}{2}$ B. $\frac{\sqrt{5}-1}{4}$ C. $\frac{\sqrt{5}+1}{4}$ D. $\frac{\sqrt{5}+1}{2}$

Approach: Use isosceles triangle properties ($AB = AC \Rightarrow \angle B = \angle C = 72^\circ$), compute $\angle A = 180^\circ - 144^\circ = 36^\circ$, then evaluate $\cos 36^\circ$.



Scene 1: The diagram draws an acute scalene triangle with visibly unequal sides, violating the given $AB = AC$ constraint. The 72° angles are not rendered at vertices B and C.

Step 2: $\angle A + 72^\circ + 72^\circ = 180^\circ \Rightarrow \angle A = 36^\circ$.



Scene 2: The highlighted triangle still shows inconsistent proportions. The isosceles constraint ($AB = AC$) is visually absent, making the diagram misleading.

Step 3: $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$

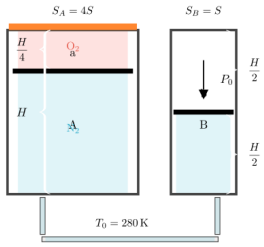
Final Answer: C ✓ (textually correct)

Failure mode (Diagram–Problem Alignment): Although the textual reasoning is correct, the rendered diagrams do not reflect the isosceles structure of the triangle—a fundamental geometric constraint of the problem. A student relying on the diagrams would form incorrect spatial intuitions.

Low-quality example 8 (Diagram–Problem Alignment failure). Diagrams fail to represent the isosceles triangle ($AB = AC$) and the 72° base angles. Despite correct textual reasoning, the visual representations are geometrically inconsistent with the problem statement, undermining pedagogical effectiveness.

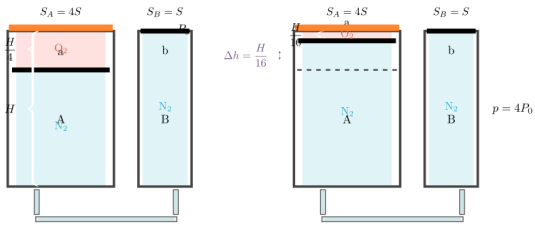
Problem: Two connected vertical cylinders A (closed top, diameter $2d_B$, cross-section $4S$) and B (open top, diameter d_B, cross-section S), both height H. Piston a in A is $H/4$ from the top; piston b in B is at mid-height. N_2 fills below both pistons; O_2 fills above piston a. (1) Heat N_2 until piston b reaches the top: find N_2 temperature. (2) Continue heating until piston a rises $H/16$: find O_2 pressure.

Approach: Stage I: N_2 expands at constant pressure P_0 (piston b free); O_2 isothermal (constant volume). Stage II: N_2 volume in B fixed; O_2 compressed isothermally by Boyle's law.

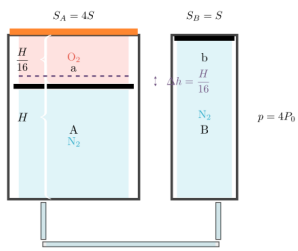


Scene 1: The height label below piston a in cylinder A is marked H instead of the correct $3H/4$, contradicting the given initial geometry and making the diagram inconsistent with the problem.

Step 2: Stage I—N_2 volume changes from $V_{N0} = 4S \cdot \frac{3H}{4} + S \cdot \frac{H}{2} = \frac{7SH}{2}$ to $V_{N1} = \frac{3SH}{4} \cdot 4S + SH = 4SH$. By $pV \propto T$: $T_1 = T_0 \cdot \frac{4SH}{7SH/2} = \frac{8}{7}T_0$.



Scene 2: After piston b reaches the top, the diagram still carries the incorrect label from Scene 1, propagating the error into Stage II.



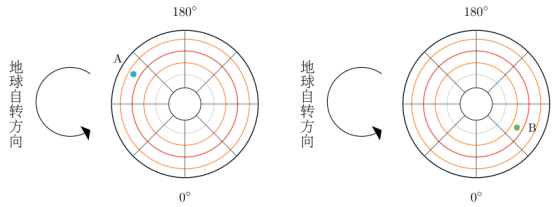
Scene 3: Stage II diagram showing piston a risen by $H/16$; layout partially corrected but inconsistent with Scene 1.

Failure mode (Element Layout Quality / Diagram–Problem Alignment): The mislabeled height H (should be $3H/4$) in Scene 1 violates the problem's geometric constraints. A student using the diagram to set up equations would obtain wrong initial volumes and incorrect answers.

Low-quality example 9 (Diagram–Problem Alignment / Element Layout Quality failure). A critical geometric parameter ($3H/4$, the nitrogen column height in cylinder A) is mislabeled as H in Scene 1 and Scene 2. The textual solution is correct, but the diagram directly contradicts the problem setup.

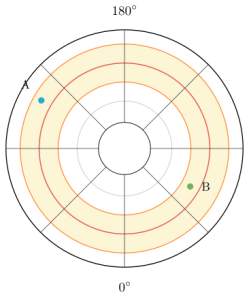
Problem: From a polar-view diagram of Earth with latitude circles and longitude markings, determine which statement is correct: (A) Point A receives direct sunlight only once a year. (B) Point B may receive direct sunlight twice a year. (C) Point B is in the Southern Hemisphere. (D) A is in the Eastern Hemisphere, B is in the Western Hemisphere.

Key facts: Direct sunlight occurs only between 23.5°S and 23.5°N. Points on the tropics receive it once; points inside receive it twice. Hemisphere boundaries: equator (N/S), 20°W–160°E (E/W).



Scene 1: Polar-view diagram. The model incorrectly identifies point A as lying outside the tropics (mid-latitude), concluding A receives no direct sunlight—in fact, A is on the equator and receives direct sunlight twice a year.

Flawed reasoning: The model misreads A as a mid-latitude point, eliminating option A with wrong logic. It then claims the diagram lacks explicit $0^\circ/180^\circ$ longitude markings to determine the E/W hemisphere—but the diagram does contain computable longitude references, making option D verifiable. The model eliminates options A, C, D through incorrect intermediate steps and selects B by elimination.



Scene 2: Hemisphere analysis diagram. The longitude reference lines drawn are inconsistent with the original diagram's markings, reflecting the model's failure to correctly read the given coordinate information.

Final Answer: B ✓ (coincidentally correct)

Failure mode (Logical Coherence / Diagram–Problem Alignment): Correct final answer reached through systematically flawed reasoning—wrong latitude identification for A, and incorrect claim about missing longitude markers. This is a "right answer, wrong method" failure that would mislead students.

Low-quality example 10 (Logical Coherence / Diagram–Problem Alignment failure). The model arrives at the correct answer (B) through two critical reasoning errors: misidentifying point A's latitude zone, and falsely claiming the diagram lacks longitude references. Such reasoning, if adopted by students, instills incorrect map-reading habits.

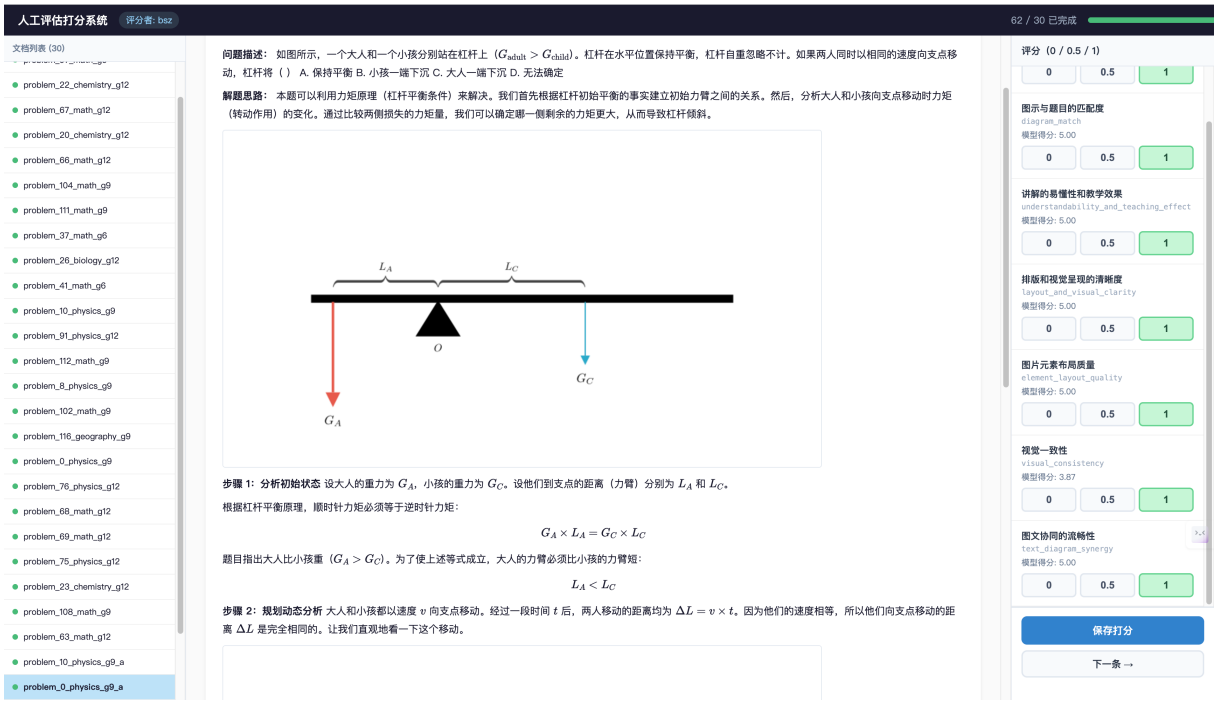


Figure 7: Screenshot of the human annotation website used for expert evaluation. Raters scored 7 dimensions on a 3-level scale after reviewing the problem, gold solution, and generated explanation with diagrams.

Model (Generator)	Gemini Judge	GPT-5 Judge
Gemini 3.0 Pro Preview	4.40	4.76
Kimi-K2.5	4.07	4.20
Claude Sonnet 4.5	3.12	3.43
GPT-5	3.11	3.21
Qwen 3.5-35B	2.98	2.67

Table 8: Overall scores (0–5) assigned by Gemini 3.0 Pro Preview and GPT-5 as judge models, evaluated on the same 50 problems per generator model. Model ranking is consistent across both judges.